PRACTICAL NUMBER: 1

AIM: Array based implementation of stacks or queues.

***PROGRAM*:**

FOR STACK:(1A)

```
//sheikh wasimuddin 60 cseiot 23071044//
#include<stdio.h>
#include<stdlib.h>
int max= 5;
int stack[5];
int top= -1;
void push(int x){
	if(top==max-1){
		printf("Overflow");
	}
	else{
		top++;
		stack[top]= x;
	}
}
void pop(void){
	if(top==-1){
		printf("stack overflow");
```

```
        }
        else{
                printf("element %d is popped from ", stack[top]);
        top--;
        }
}
void peep(void){
        if(top==-1){
                printf("underflow");
        }
        else{
                printf("element at top is %d", stack[top]);
        }
}
void display(){
int i;
for(i=top; i>=0; i--){
        printf("%d ", stack[i]);
}
}
int main(){
        int choice;
        int e;
        while(choice!=5){
```

```
    printf("stack\n");
printf("1: push element\n2: pop element\n3: peek the
stack\n4:display stack\n5: exit\nEnter the choice: ");
scanf("%d", &choice);
switch(choice){
        case 1:
                printf("enter element:");
                scanf("%d", &e);
                push(e);
                break;
        case 2:
                pop();
                break;
        case 3:
                peep();
                printf("\n");
                break;
        case 4:
                display();
                printf("\n");
                break;
        case 5:
                printf("exited\n");
                printf("\n");
```

```
                break;
            default:
                printf("wrong choice\n\n");
                break;
        }}
return 0;
}
```

OUTPUT:

main.c

```c
//sheikh wasimuddin 60 cseiot 23071044
#include<stdio.h>
#include<stdlib.h>
int max= 5;
int stack[5];
int top= -1;
void push(int x){
    if(top==max-1){
        printf("Overflow");
    }
    else{
        top++;
        stack[top]= x;
    }
}
void pop(void){
    if(top==-1){
        printf("stack overflow");
    }
    else{
        printf("element %d is popped from ", stack[top]);
    top--;
    }
}
void peep(void){
    if(top==-1){
        printf("underflow");
    }
    else{
        printf("element at top is %d", stack[top]);
    }
}
void display(){
int i;
for(i=top; i>=0; i--){
```

```
stack
1: push element
2: pop element
3: peek the stack
4: display stack
5: exit
Enter the choice: 1
enter element:11
stack
1: push element
2: pop element
3: peek the stack
4: display stack
5: exit
Enter the choice: 1
enter element:22
stack
1: push element
2: pop element
3: peek the stack
4: display stack
5: exit
Enter the choice: 1
enter element:33
stack
1: push element
2: pop element
3: peek the stack
4: display stack
5: exit
Enter the choice: 4
33 22 11
stack
1: push element
2: pop element
3: peek the stack
4: display stack
5: exit
Enter the choice: 3
element at top is 33
stack
```

main.c

```c
//sheikh wasimuddin 60 cseiot 23071044
#include<stdio.h>
#include<stdlib.h>
int max= 5;
int stack[5];
int top= -1;
void push(int x){
    if(top==max-1){
        printf("Overflow");
    }
    else{
        top++;
        stack[top]= x;
    }
}
void pop(void){
    if(top==-1){
        printf("stack overflow");
    }
    else{
        printf("element %d is popped from ", stack[top]);
    top--;
    }
}
void peep(void){
    if(top==-1){
        printf("underflow");
    }
    else{
        printf("element at top is %d", stack[top]);
    }
}
void display(){
int i;
for(i=top; i>=0; i--){
```

```
3: peek the stack
4: display stack
5: exit
Enter the choice: 4
33 22 11
stack
1: push element
2: pop element
3: peek the stack
4: display stack
5: exit
Enter the choice: 3
element at top is 33
stack
1: push element
2: pop element
3: peek the stack
4: display stack
5: exit
Enter the choice: 2
element 33 is popped from stack
1: push element
2: pop element
3: peek the stack
4: display stack
5: exit
Enter the choice: 4
22 11
stack
1: push element
2: pop element
3: peek the stack
4: display stack
5: exit
Enter the choice: 5
exited

...Program finished with exit code 0
Press ENTER to exit console.
```

FOR QUEUE: (1B)

//sheikh wasimuddin 60 cseiot 23071044//

```c
#include<stdio.h>

#include<stdlib.h>

int max= 5;

int queue[5];

int rear = -1;
```

```
int front = -1;

void enqueue(int x){
	if(rear==max-1){
		printf("overflow");
	}
	else if(front==-1 && rear==-1){
		front= rear= 0;
		queue[rear]= x;
	}
	else{
		rear++;
		queue[rear]= x;
	}
}

void dequeue(void){
	if(front==-1 && rear==-1){
		printf("underflow");
	}
	else if(front==rear){
		printf("element dequeued is %d", queue[front]);
		front= rear= -1;
	}
```

```
        else{
                printf("dequeue %d", queue[front]);
                front++;
        }
}

void display(void){
        int i;
        for(i=front; i<=rear; i++){
                printf("%d ", queue[i]);
        }
}

int main(){
        int choice;
        int x;
        while(choice!=4){
        printf("queue\n");
        printf("1: Enqueue \n2: Dequeue\n3: Display\n4: exit\nEnter
        choice:");
        scanf("%d", &choice);
        switch(choice){
                case 1:
                        printf("enter element to insert:\n");
```

```
                scanf("%d", &x);
                enqueue(x);
                break;
            case 2:
                dequeue();
                break;
            case 3:
                display();
                break;
            case 4:
                printf("exited\n");
                break;
            default:
                printf("wrong choice");
                break;
        }
        }
        return 0;
}
```

OUTPUT:

main.c

```c
//sheikh wasimuddin 60 cseiot 23071044
#include<stdio.h>
#include<stdlib.h>
int max= 5;
int queue[5];
int rear = -1;
int front = -1;

void enqueue(int x){
    if(rear==max-1){
        printf("overflow");
    }
    else if(front==-1 && rear==-1){
        front= rear= 0;
        queue[rear]= x;
    }
    else{
        rear++;
        queue[rear]= x;
    }
}

void dequeue(void){
    if(front==-1 && rear==-1){
        printf("underflow");
    }
    else if(front==rear){
        printf("element dequeued is %d", queue[front]);
        front= rear= -1;
    }
    else{
        printf("dequeue %d", queue[front]);
        front++;
    }
}
```

```
1: Enqueue
2: Dequeue
3: Display
4: exit
Enter choice:1
enter element to insert:
33
queue
1: Enqueue
2: Dequeue
3: Display
4: exit
Enter choice:1
enter element to insert:
22
queue
1: Enqueue
2: Dequeue
3: Display
4: exit
Enter choice:1
enter element to insert:
11
queue
1: Enqueue
2: Dequeue
3: Display
4: exit
Enter choice:3
33 22 11 queue
1: Enqueue
2: Dequeue
3: Display
4: exit
Enter choice:2
dequeue 33queue
1: Enqueue
2: Dequeue
3: Display
4: exit
Enter choice:4
exited
```

CONCLUSION: Array based implementation of stacks or queues is performed successfully.